Discovering Leitmotifs in Multidimensional Time Series

Research track: Time Series Data III

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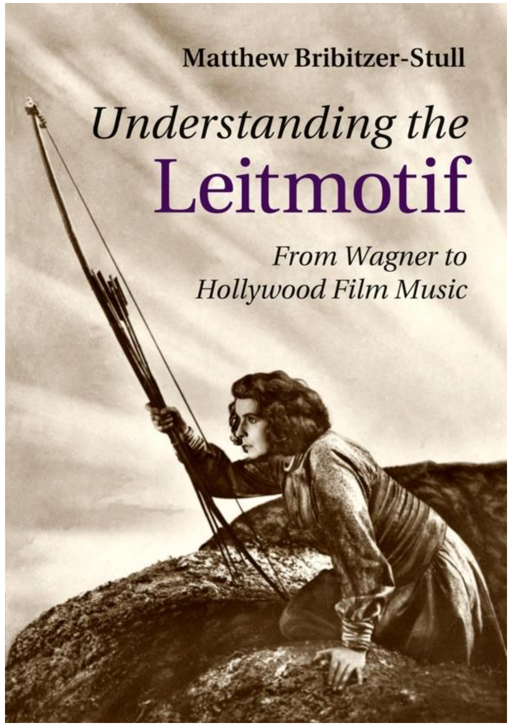
Leitmotifs



- A leitmotif (leading motif) is a recurring theme or „motif“

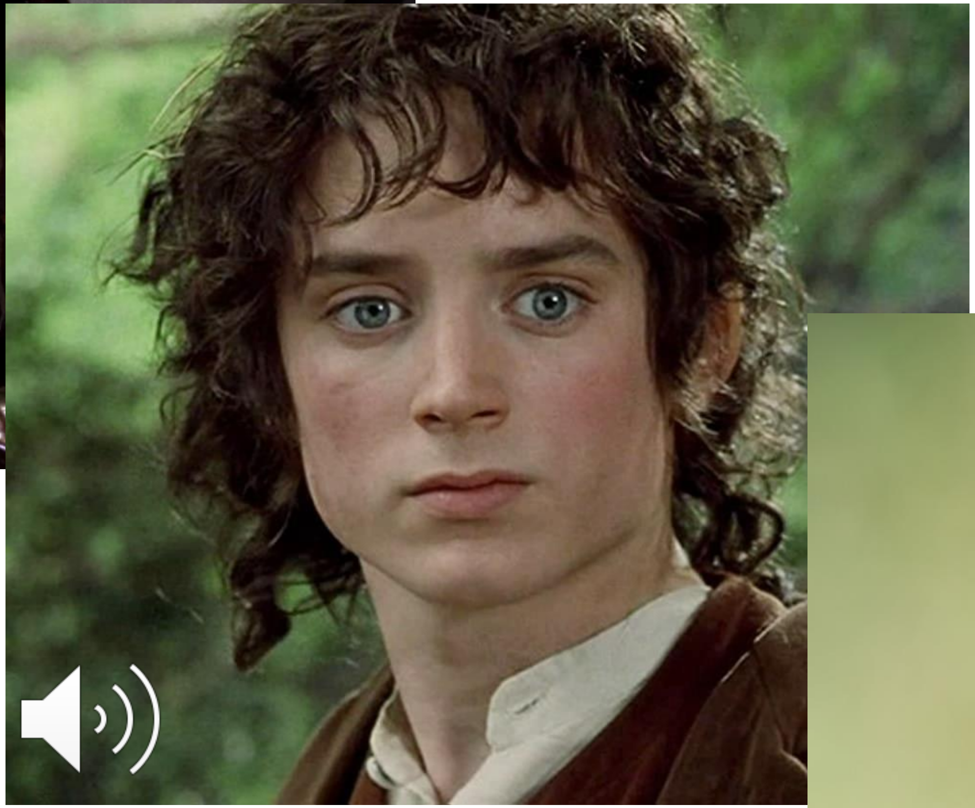
- It carries symbolic significance in all forms of art, e.g. literature, movies, and music

- Humans associate leitmotifs with meaning, which enhances narrative cohesion and establishes emotional connections with the audience
  - **Richard Wagner** used these as a compositional technique in his operas

Ride of the Valkyries

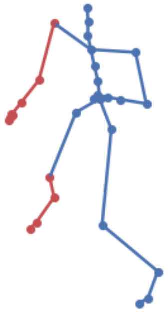
John Williams – Imperial March

Howard Shore – The Shire

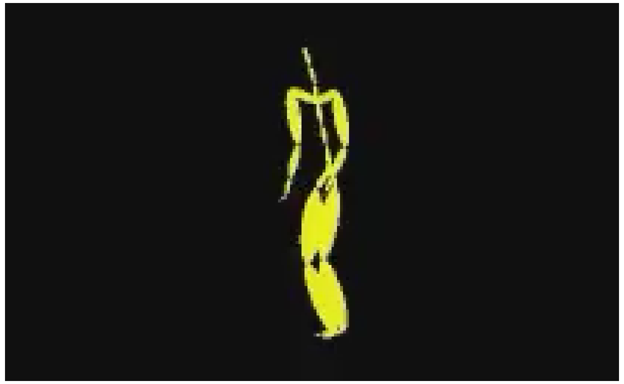


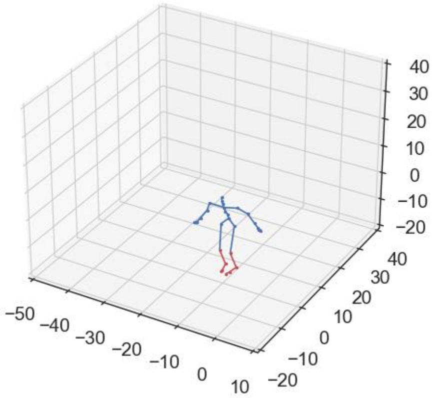
Common Starling

Motion Capture: Boxing Routine



Motion Capture: Charlston Dancing



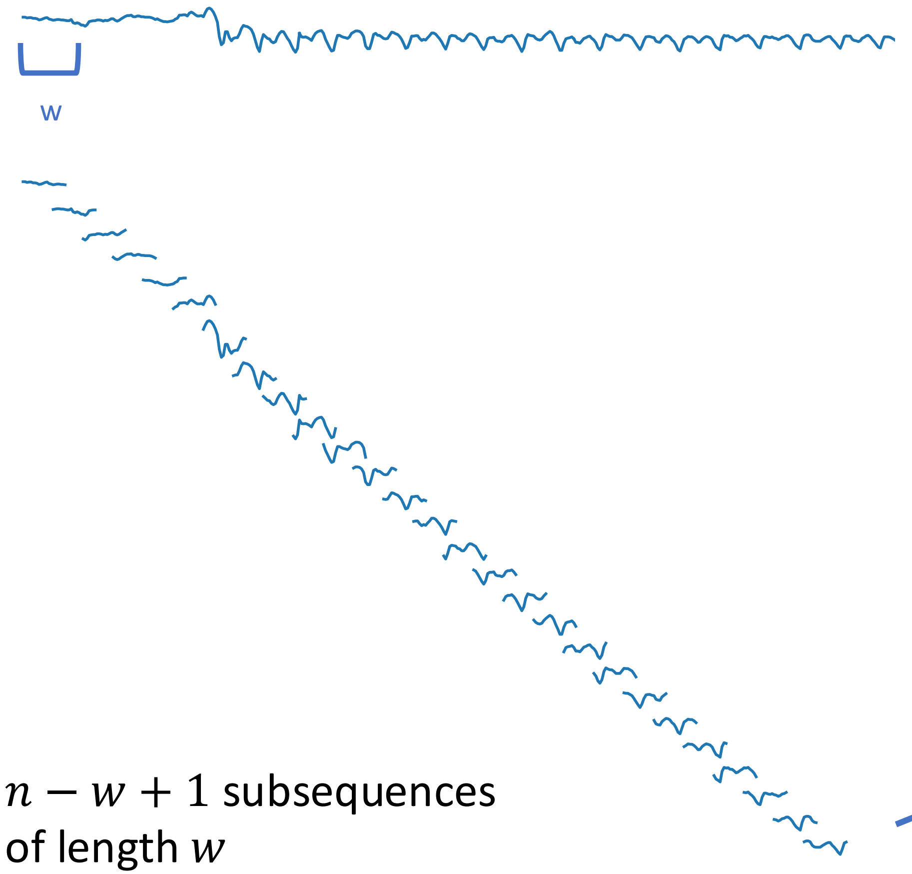


Challenging Problem

- A **leitmotif** is a **sub-dimensional motif set** of a time series with **symbolic significance** which
  - has unknown length l,
  - occurs an unknown number of times k, and
  - occurs in an unknown number of sub-dimensions f
- It is an unsupervised and mathematically ill-defined problem of high complexity

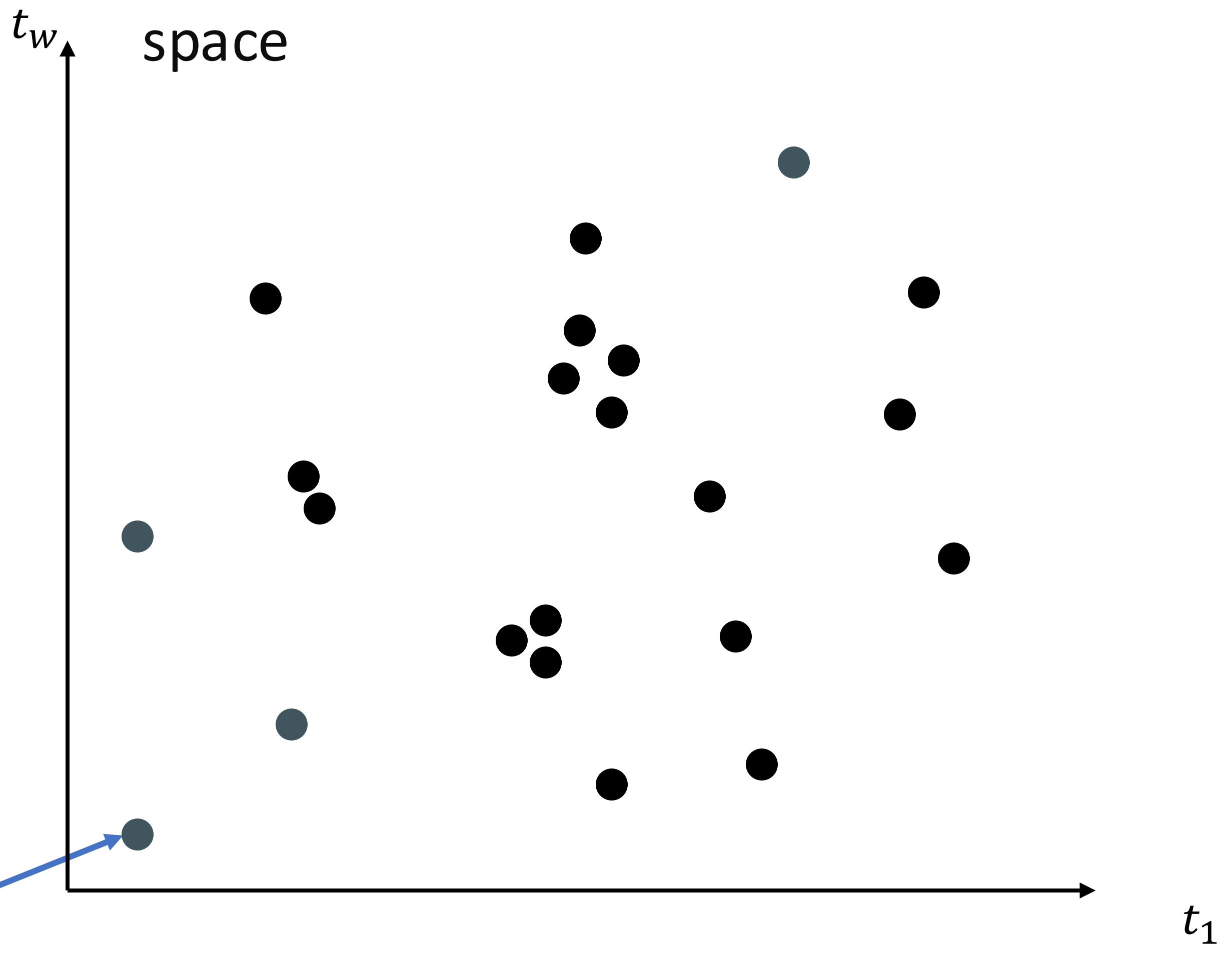
Background: High-Dimensional Vector Space

We use a sliding window to extract w-length subsequences



$n - w + 1$ subsequences of length w

... and interpret these as points in w-dim space

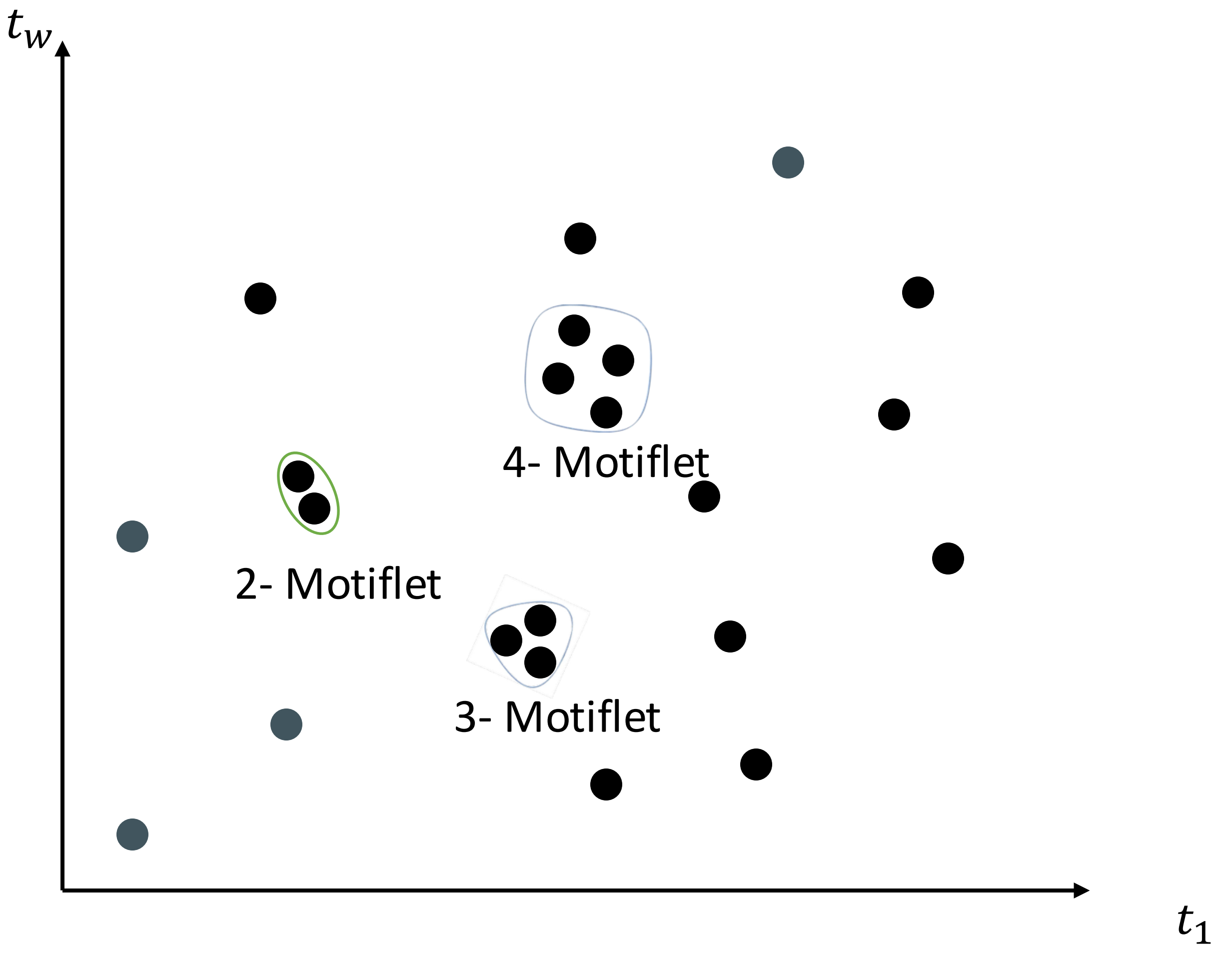


$n - w + 1$ points of dim w

Motiflets – VLDB`24

- Goal: Find smallest sets of similar points

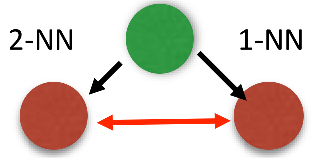
- Motiflets: find k-frequent motif sets with **smallest pairwise distance (aka extent)**



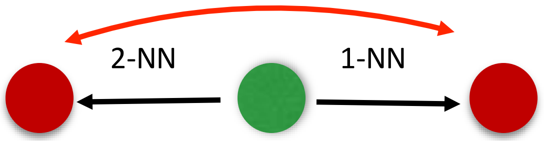
Motiflets – Algorithm

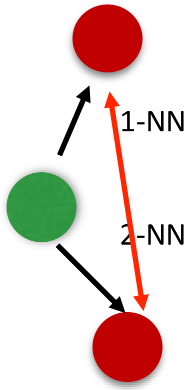
Input: freq. *k* and length *l*

1. Perform a k-NN search around each of $O(n)$ subsequences *(red dots)*
2. For each candidate, compute maximum over pairwise distances (*“extent”*)



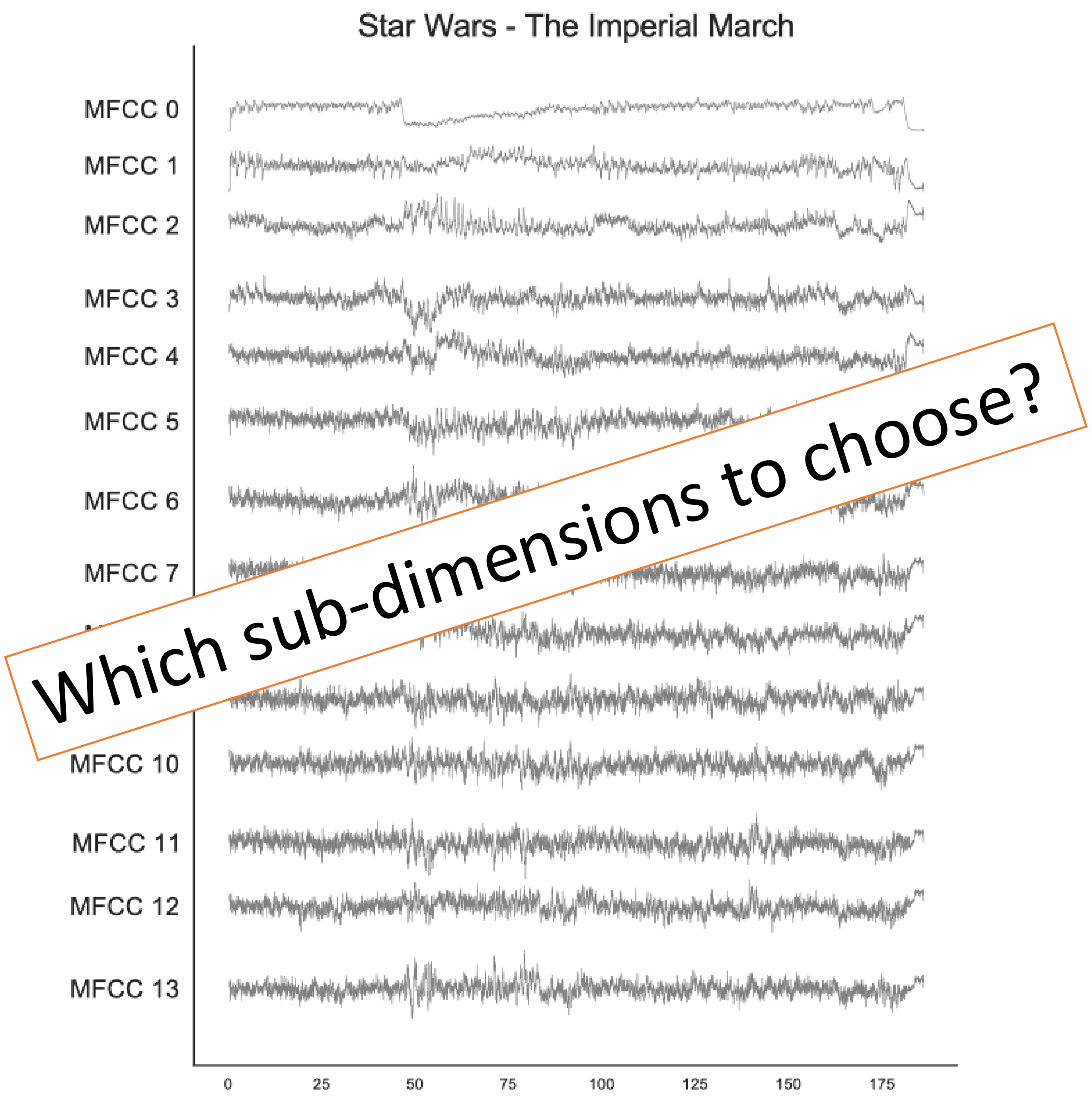
smallest extent





3. Return: Minimum over all candidate sets

- Complexity: $O(k \cdot n^2)$
- **But: Only applicable for one-dimensional data**


Star Wars - The Imperial March
MFCC 0
MFCC 1
MFCC 2
MFCC 3
MFCC 4
MFCC 5
MFCC 6
MFCC 7
MFCC 10
MFCC 11
MFCC 12
MFCC 13
0
25
50
75
100
125
150
175
Which sub-dimensions to choose?

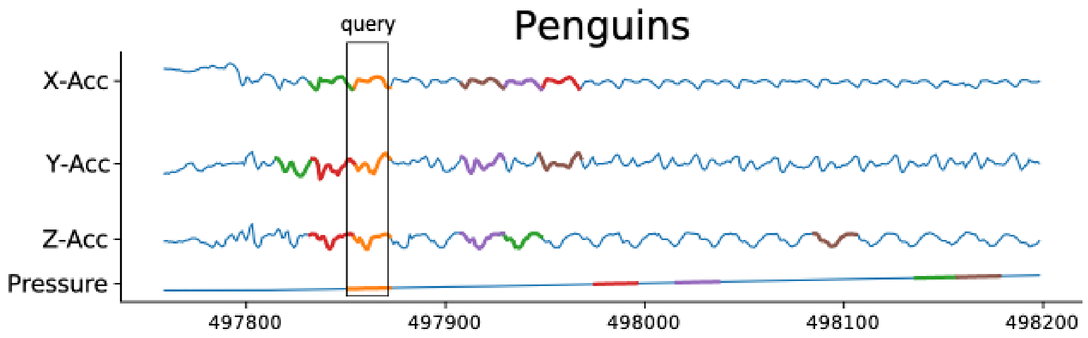
Subdimensional Motif Sets

- Searching optimal Motif Sets along sub-dimensions is NP-hard

- We must enumerate all subsets of f out of d-dims
  - There are $\binom{d}{f}$ – many combinations

- We propose an approximate solution: LAMA

LAMA



- For each a query subsequence:
  1. perform k-NN search along each dimension
  2. sort dimensions by kth NN distance
  3. take f smallest values
  4. use *offsets* from dimension with smallest distance
  5. compute *extent* as sum over f dims

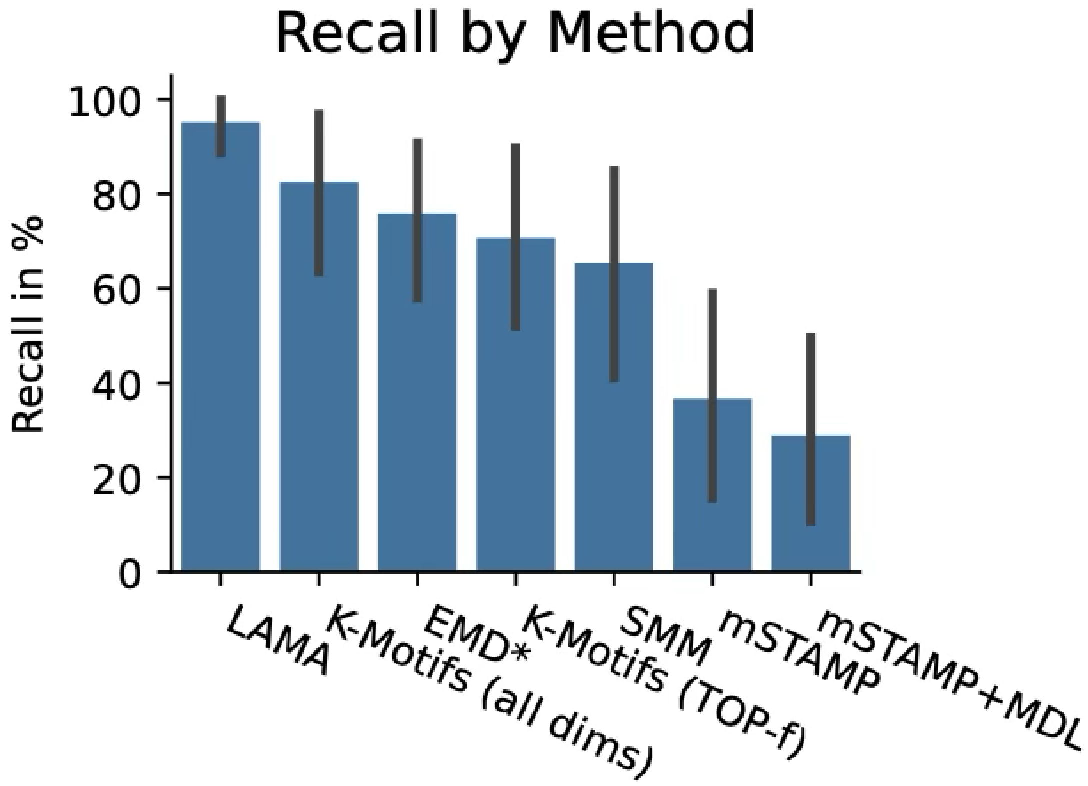
- Minimize extent over all f-dim candidates
- Worst / Best Case: $O(d\ k\ n^2)$ / $O(d\ n^2)$

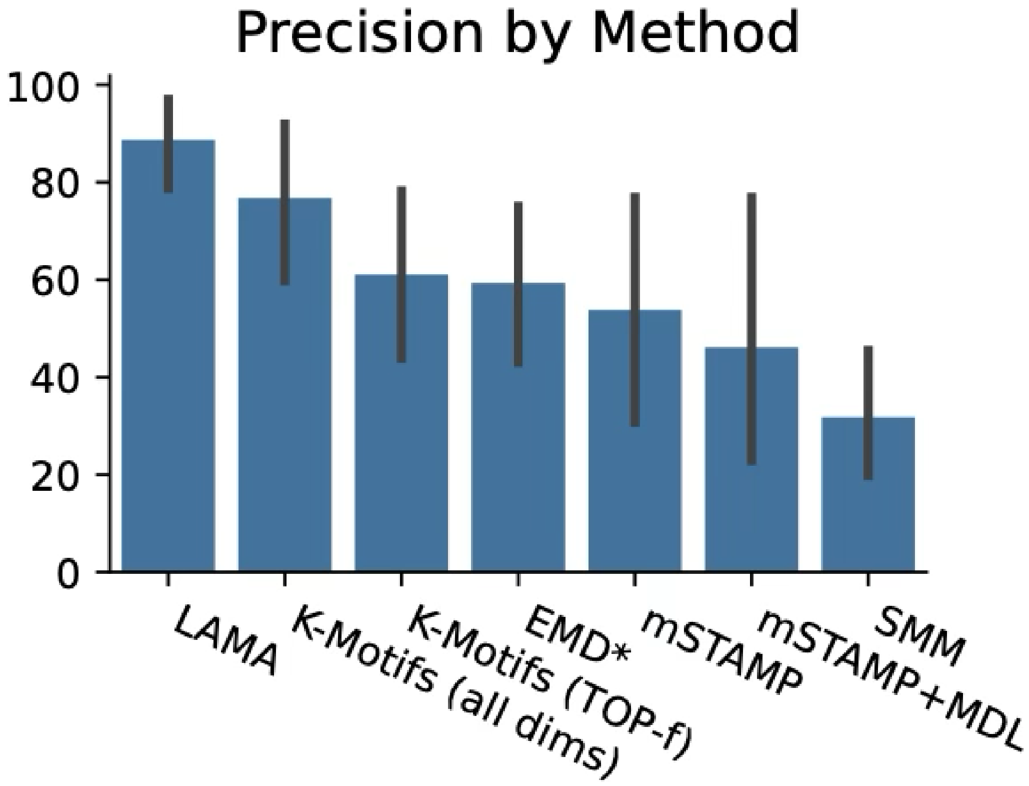
Experiments

- We evaluated LAMA on 14 real-world use cases
- Four competitors
  - **SMM**: subdimension motif set discovery
  - **mSTAMP**: subdimensional pair motif discovery
  - **EMD**: PCA for channel reduction, followed by univarite motif ser discovery
  - **K-Motifs**: aggregate distances for first f or all channels
- Each real-world use case has ground truth leitmotifs manually annotated
  - We can measure Precision/Recall

Use Case	Category	Length	Dim.	GT
Charleston [9]	Motion Capture	506	93	3
Boxing [9]	Motion Capture	4840	93	10
Swordplay [9]	Motion Capture	2251	93	6
Basketball [9]	Motion Capture	721	93	5
LOTR - The Shire	Soundtrack	6487	20	4
SW - The Imperial March	Soundtrack	8015	20	5
RS - Paint it black	Pop Music	9744	20	10
Linkin Park - Numb	Pop Music	8018	20	5
Linkin P. - What I've Done	Pop Music	8932	20	6
Queen - Under Pressure	Pop Music	9305	20	16
Vanilla Ice - Ice Ice Baby	Pop Music	11695	20	20
Starling [5]	Wildlife Rec.	2839	20	4
Physiodata [24]	Wearable Sensors	5526	5	20
Bitcoin Halving	Crypto/Stocks	3591	7	3

Lama outperforms competitors

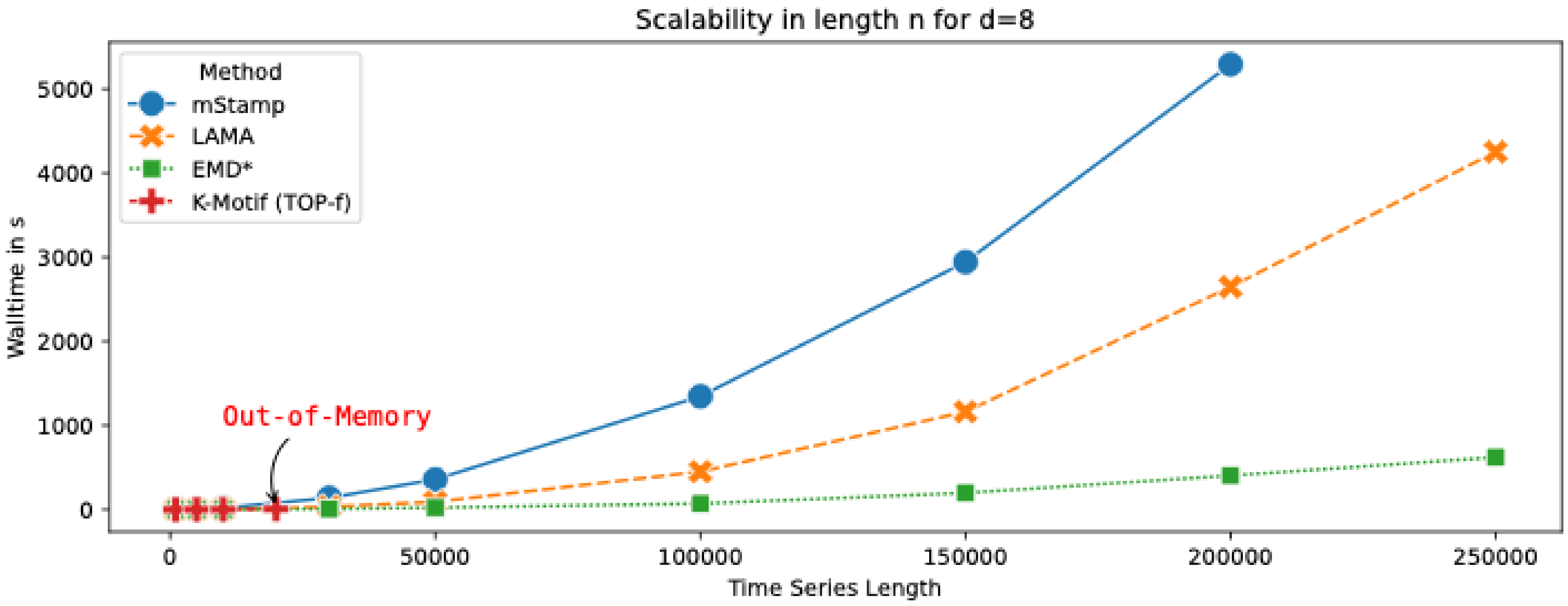




Most occurrences are found

If reported, it is correct

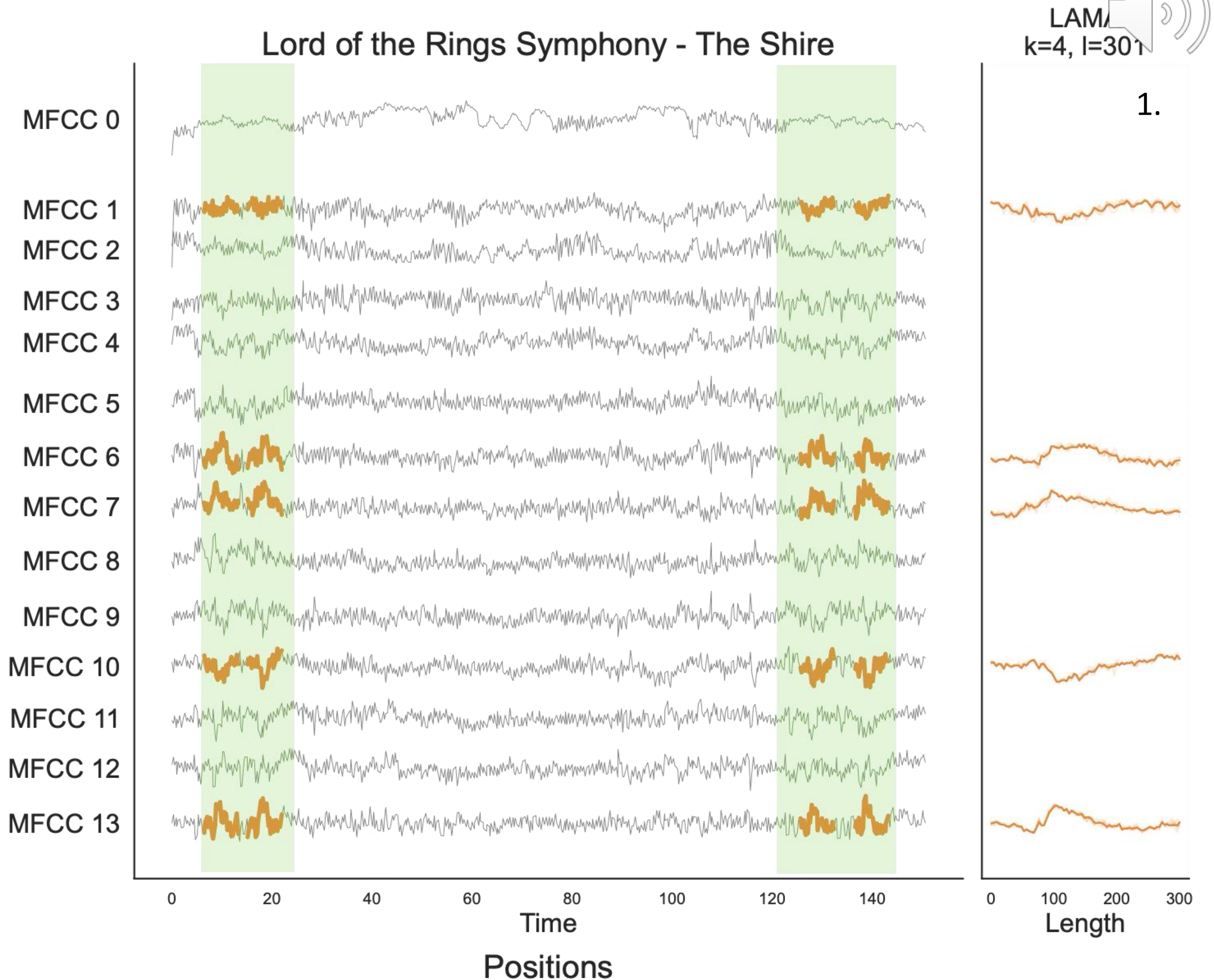
Scalability



LAMA scales better than most methods but EMD

Use Case – LOTR

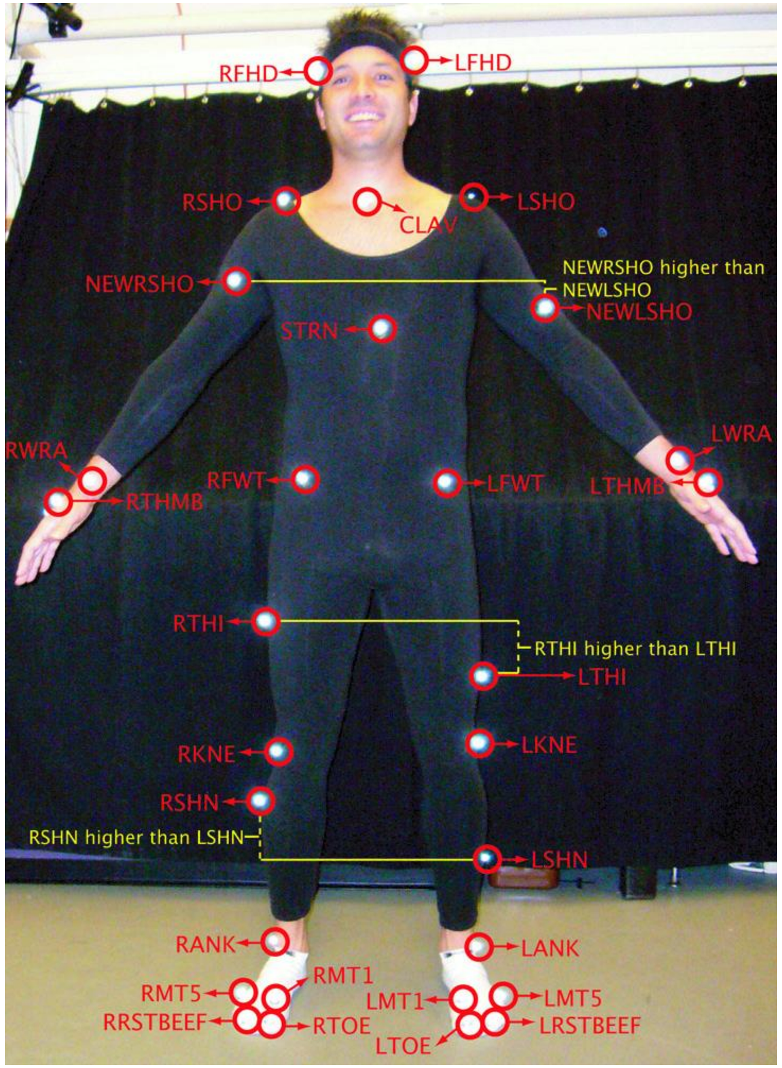
- The Shire was arranged by Howard Shore for Lord of the Rings
- It opens and ends with the Hobbits’ leitmotif, played by a solo tin whistle
  - It manifests in a distinct pattern in several, but not all dimensions



Tin Whistle playing at ~6 sec

Use Case – Motion Capture

- Data recorded by the the CMU Database
  - http://mocap.cs.cmu.edu/



Use Case – Boxing

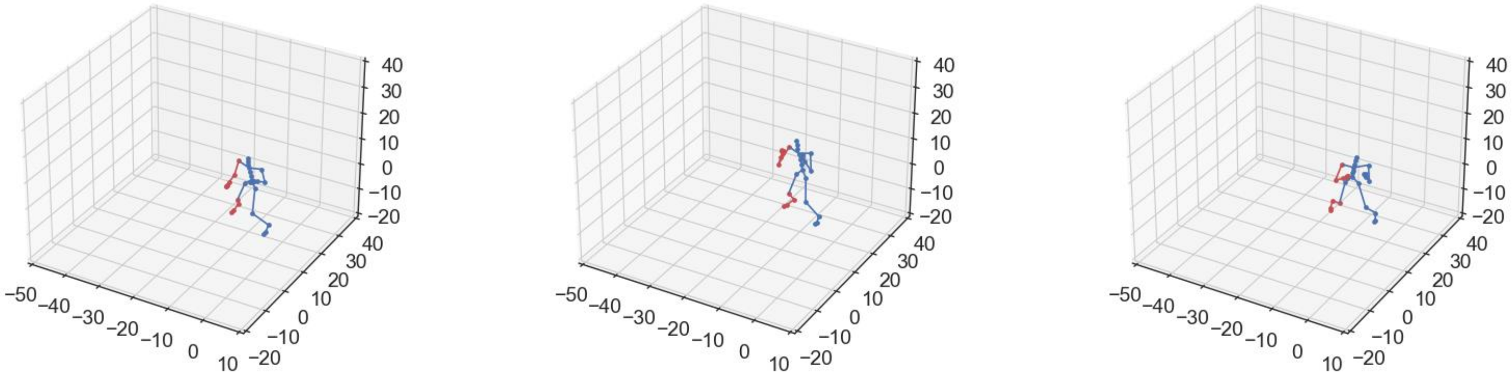
- We looked into boxing motion capture, of 41 sec at 120 frames per second

- We concentrate on right body joints:

  'rclavicle', 'rhumerus‘,
  'rradius', 'rwrist‘,
  'rhand', 'rfingers', 'rthumb‘,
  'rfemur', 'rtibia', 'rfoot', 'rtoes'

Use Case – Motion Capture

- We found 10 repeats of the leitmotif of length 0.8 sec
- It is the punching motion
- Here, showing only the first three

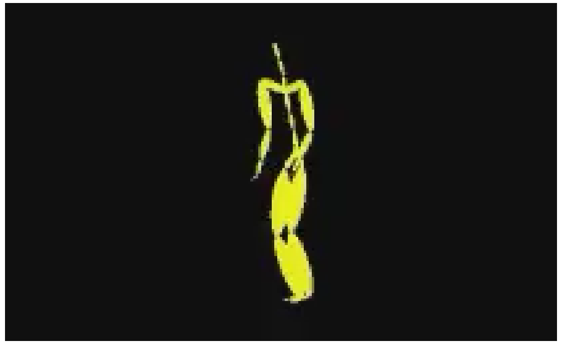


20’s Charleston

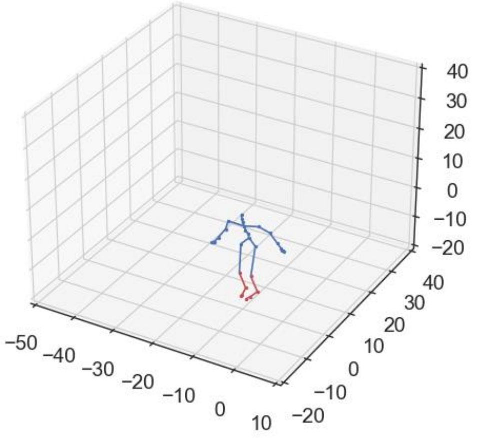
- Originated in Charleston South Carolina
- In Europe, the dance became known mainly through the singer, dancer and actress Josephine Baker

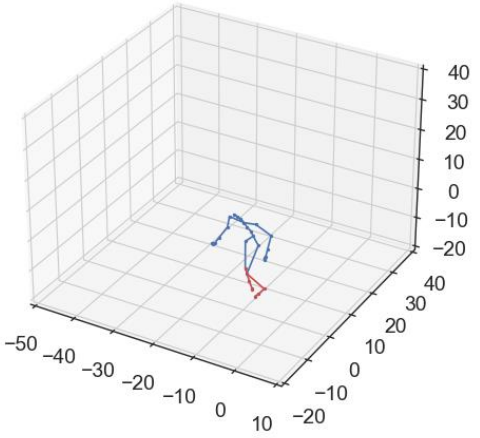
20’s Charleston

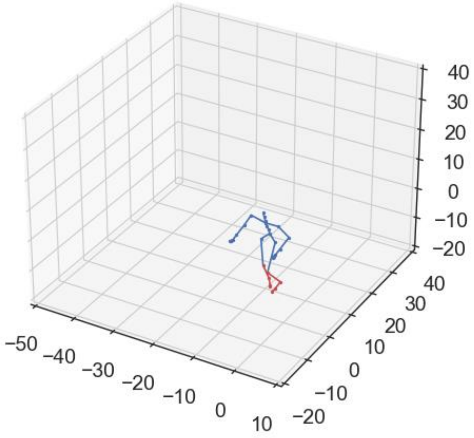
- We use LAMA to find the basic step / footwork from motion data of a Charleston dancing routine
- The used routine contains 3 repeats of this leitmotif
- The found leitmotif is the basic step



3 repeats







Conclusion

- We present the first annotated Leitmotif benchmark

- Our experimental evaluation shows that LAMA has superior performance in detecting semantically meaningful patterns

- Our findings demonstrate that summarizing a time series through its leitmotifs offers a profound understanding of underlying patterns

Source Codes

Thank you!

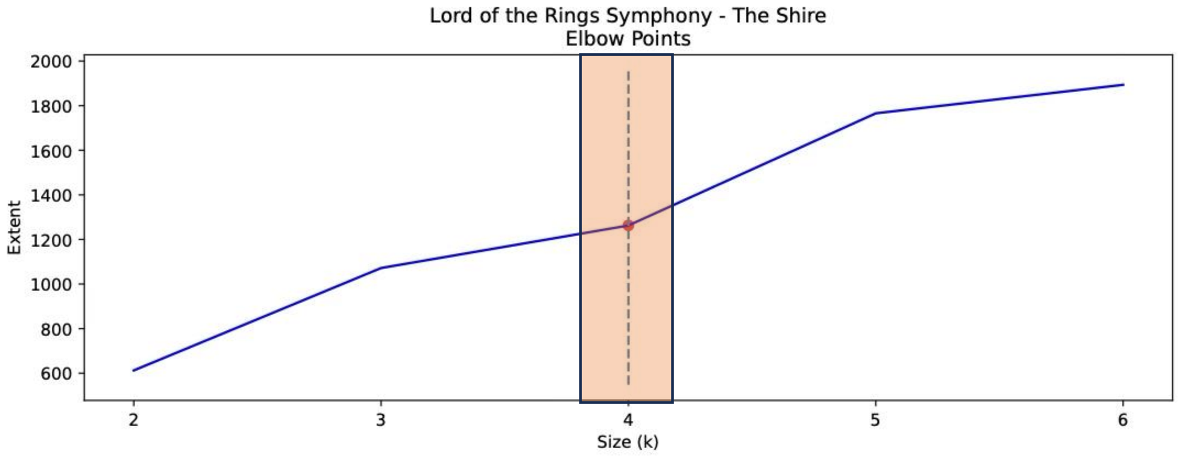
Questions?

Source Codes

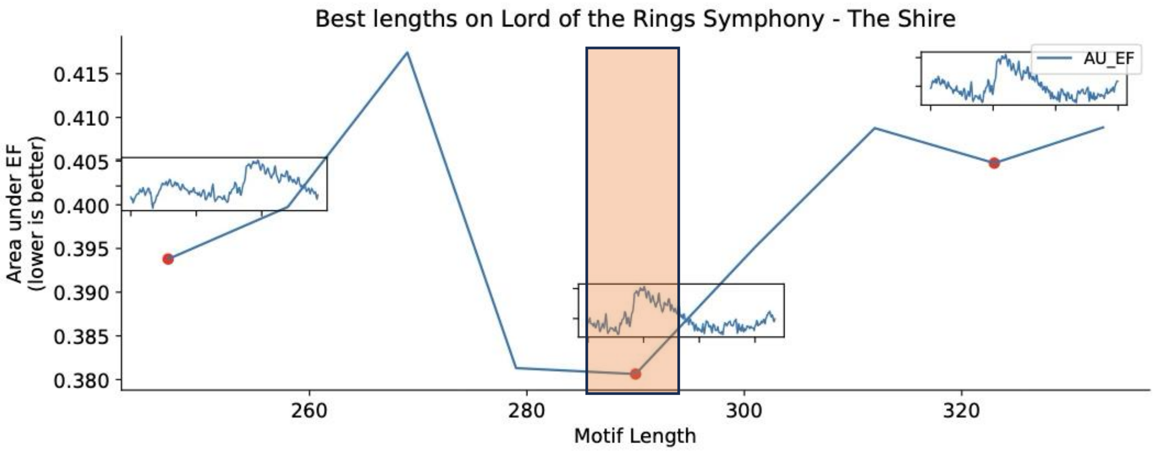
Backup Slides

Learning *k* and *l*

- Optimum at 4 repeats of roughly 7 seconds

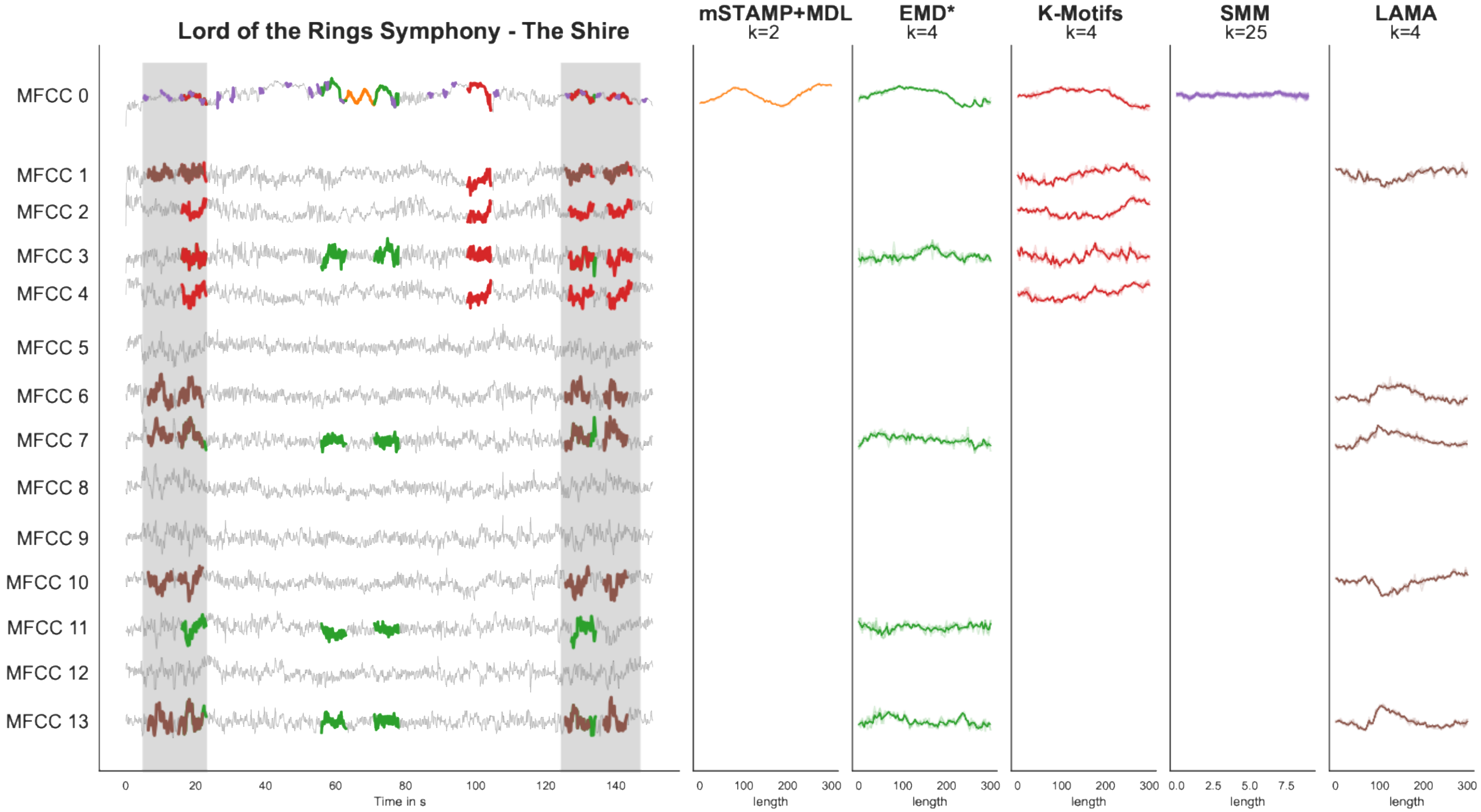


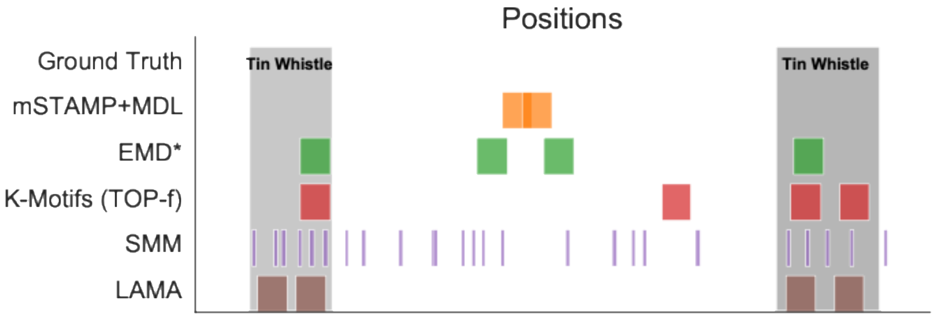
Elbow Points



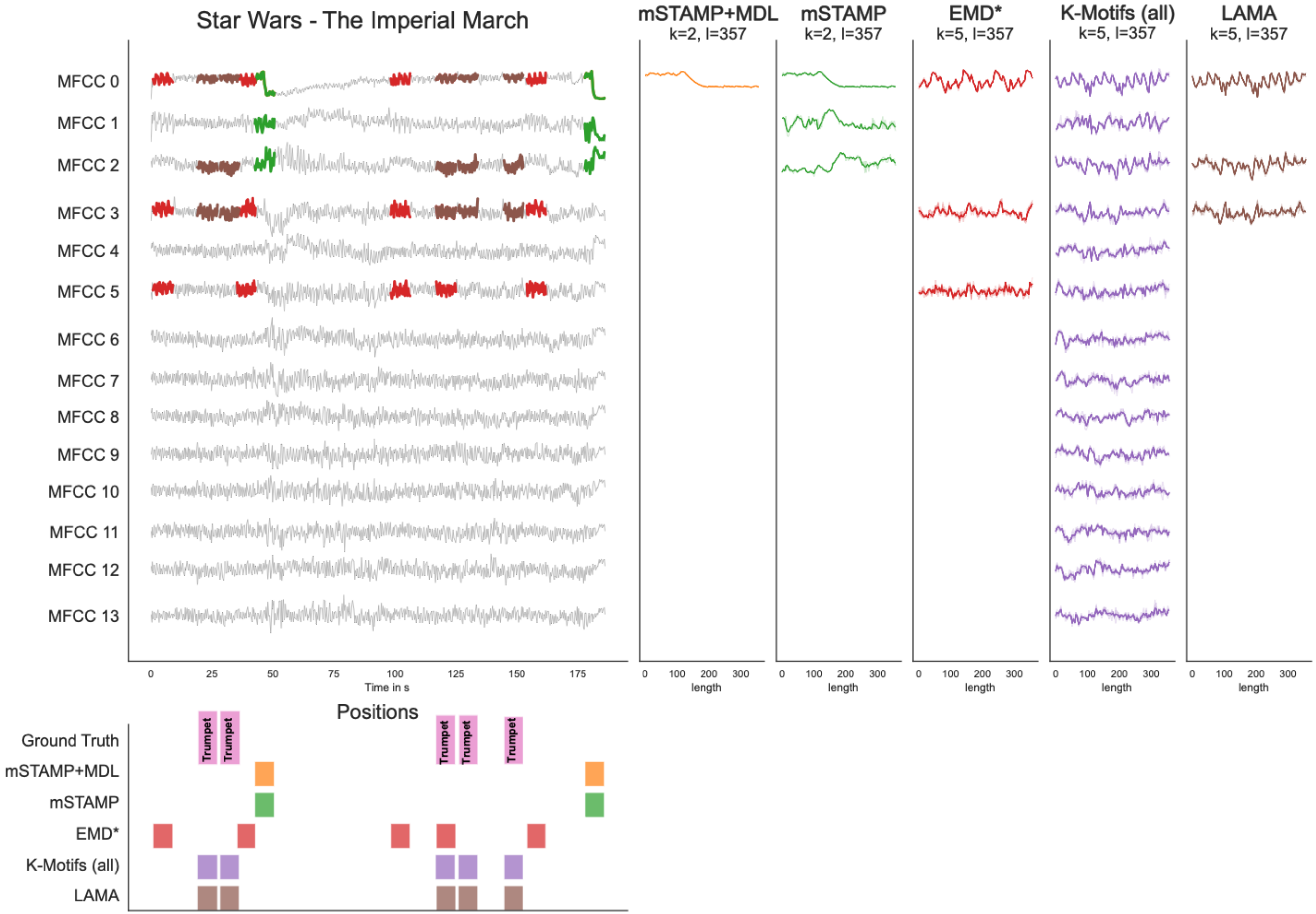
Area under the Extent Function

- Details omitted for brevity – see paper

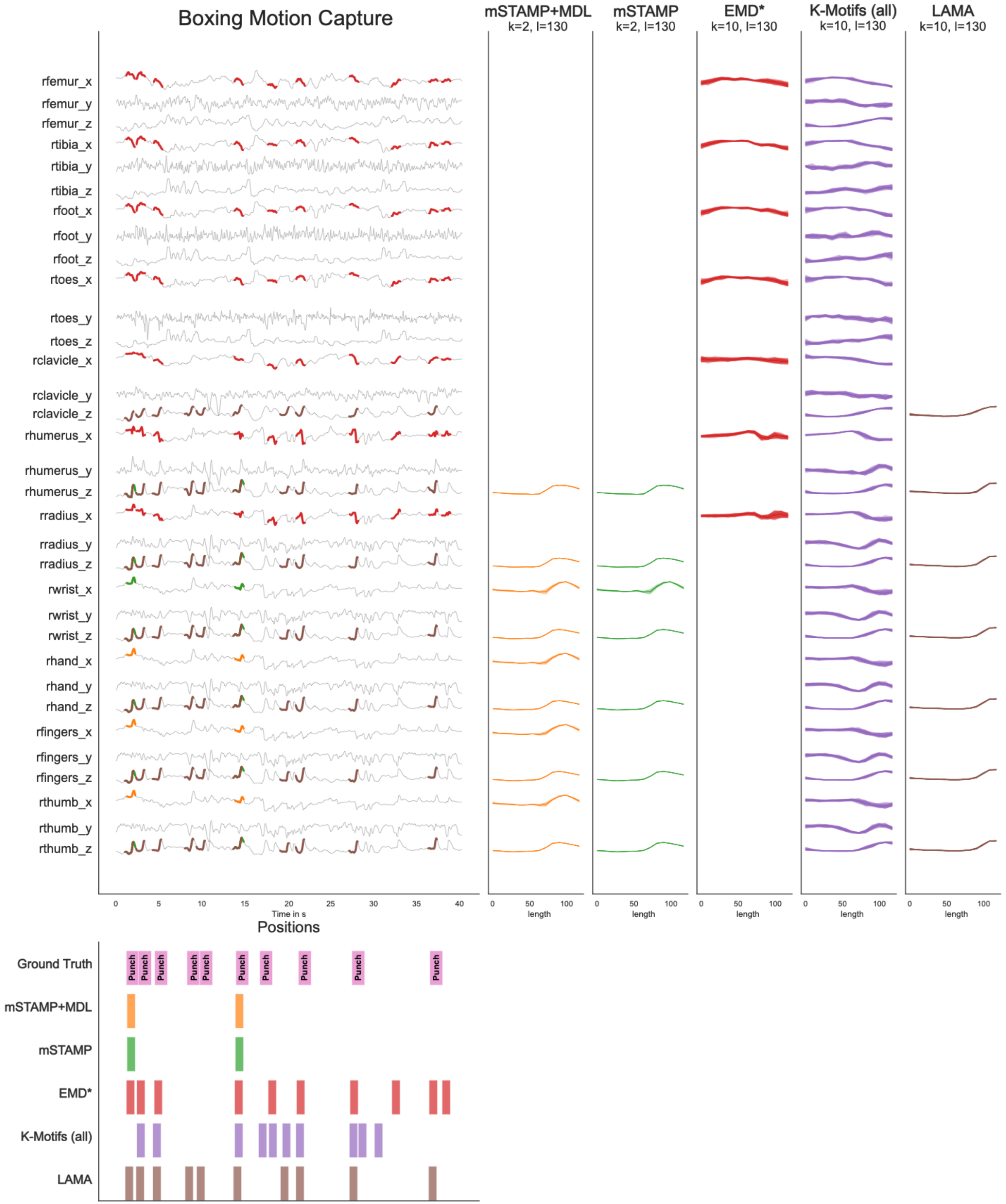




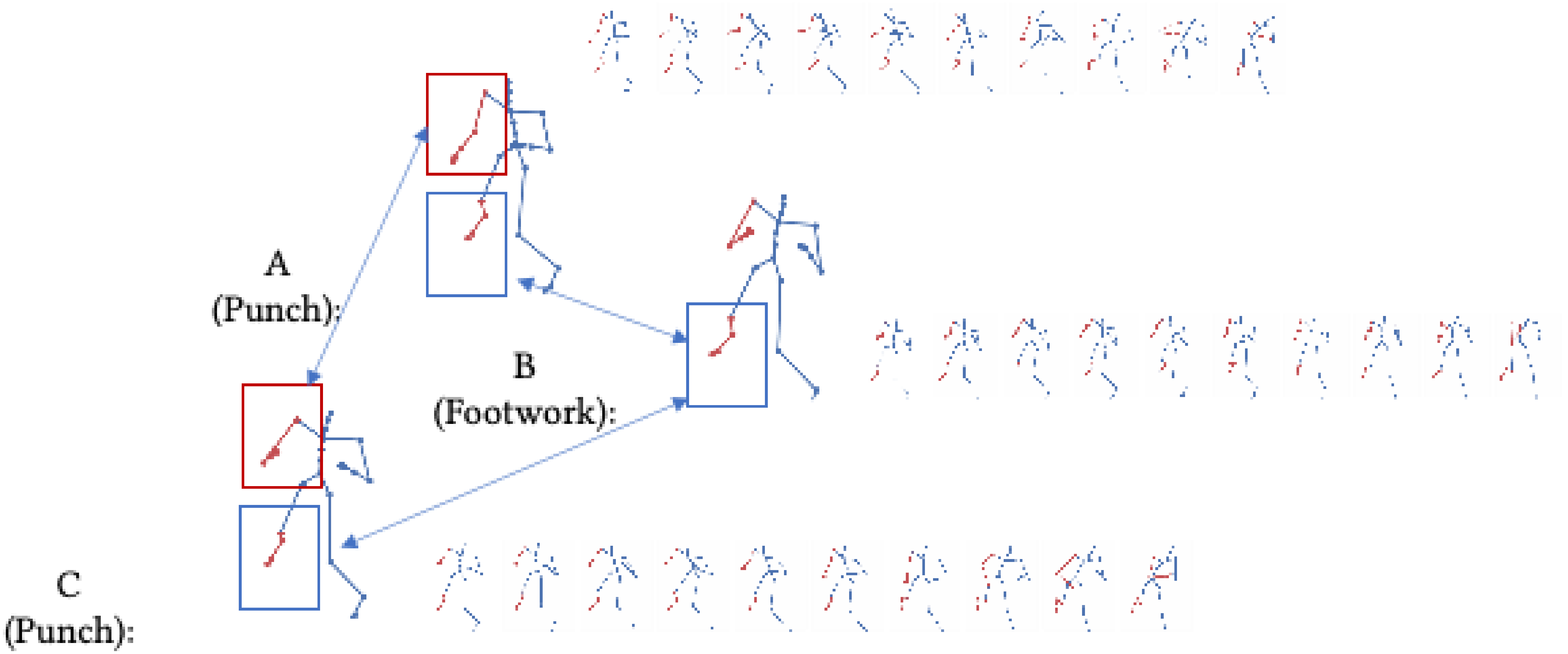
Leitmotif: Tin Whistle playing at ~6 sec



Star Wars - The Imperial March
mSTAMP+MDL
k=2, l=357
mSTAMP
k=2, l=357
EMD*
k=5, l=357
K-Motifs (all)
k=5, l=357
LAMA
k=5, l=357
MFCC 0
MFCC 1
MFCC 2
MFCC 3
MFCC 4
MFCC 5
MFCC 6
MFCC 7
MFCC 8
MFCC 9
MFCC 10
MFCC 11
MFCC 12
MFCC 13
0 25 50 75 100 125 150 175
Time in s
0 100 200 300
length
Positions
Ground Truth
mSTAMP+MDL
mSTAMP
EMD*
K-Motifs (all)
LAMA
Trumpet
Trumpet
Trumpet
Trumpet
Trumpet



Boxing Motion Capture
mSTAMP+MDL
k=2, l=130
mSTAMP
k=2, l=130
EMD*
k=10, l=130
K-Motifs (all)
k=10, l=130
LAMA
k=10, l=130
rfemur_x
rfemur_y
rfemur_z
rtibia_x
rtibia_y
rtibia_z
rfoot_x
rfoot_y
rfoot_z
rtoes_x
rtoes_y
rtoes_z
rclavicle_x
rclavicle_y
rclavicle_z
rhumerus_x
rhumerus_y
rhumerus_z
rradius_x
rradius_y
rradius_z
rwrist_x
rwrist_y
rwrist_z
rhand_x
rhand_y
rhand_z
rfingers_x
rfingers_y
rfingers_z
rthumb_x
rthumb_y
rthumb_z
0 5 10 15 20 25 30 35 40
Time in s
0 50 100
length
Positions
Ground Truth
Punch
mSTAMP+MDL
mSTAMP
EMD*
K-Motifs (all)
LAMA


A
(Punch):
B
(Footwork):
C
(Punch):